Problem – Solution Fit Template

Date	12-06-2026
Team ID	
Project Name	**Voice Based Concept Understanding Analyser**
Maximum Marks	2 Marks

Problem-Solution Fit

1.CUSTOMER SEGMENT(S) (CS) • Undergraduate & Postgraduate Students • Students preparing for Viva/Interviews • Faculty Members • Educational Institutions	**6. CUSTOMER CONSTRAINTS (CC)** • Limited faculty availability • No AI-based speech evaluation • Time constraints before exams • Difficulty identifying weak concepts	**5. AVAILABLE SOLUTIONS (AS)** • **ChatGPT/Gemini** – Good explanations, but no spoken concept evaluation. • **YouTube** – Helpful learning, but no personalised feedback. • **Traditional Viva Practice** – Useful but time-consuming and dependent on faculty availability.
2. JOBS-TO-BE-DONE / PROBLEMS (J&P) • Difficulty explaining concepts clearly • No objective evaluation of spoken answers • Lack of instant personalized feedback • Low confidence before viva/Lack of personalized learning feedback.	**9. PROBLEM ROOT CAUSE (RC)** • Existing platforms focus on written assessments. • No automated evaluation of spoken conceptual understanding. • Manual feedback is slow and inconsistent. • Students cannot accurately measure communication skills.	**7. BEHAVIOUR (BE)** Students typically: • Watch YouTube lectures. • Read textbooks and lecture notes. • Practice explaining concepts aloud. • Search Google for explanations. • Use ChatGPT for concept clarification. • Memorize instead of developing deep understanding. • Repeat explanations before interviews.
3. TRIGGERS (TR) Students use VBCUA when they: • Poor viva/interview performance • Difficulty understanding concepts • Need for self-assessment • Practice before exams.	**10. YOUR SOLUTION (SL)** **Voice-Based Concept Understanding Analyser (VBCUA)** AI-powered voice-based concept evaluation • Whisper Speech-to-Text • Sentence-BERT semantic similarity	**8. CHANNELS OF BEHAVIOUR (CH)** Online • Streamlit Web Application • ChatGPT • Google Gemini • YouTube • University LMS

	• Librosa speech analysis • Gemini AI personalized feedback • PDF report generation • Progress tracking and evaluation history	Offline • Classroom lectures • Viva examinations • Study groups • Placement training sessions
4. EMOTIONS: BEFORE / AFTER (EM) **Before** • Confused • Nervous • Underprepared **After** • Confident • Motivated • Better prepared		